What Makes a Generative Latent Space OOD-Aware? A Unified Analysis of Geometry and Density Across VAEs, GANs, and Diffusion Models

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***Abstract*—Out-of-distribution (OOD) detection is essential for machine learning systems that encounter inputs different from those observed during training. Generative models provide internal representations that may help distinguish familiar data from unfamiliar samples; however, their effectiveness depends on the properties of the learned representation and the detection method. This paper proposes a unified framework to investigate OOD detection across Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models. Rather than focusing exclusively on architectural performance, the framework examines latent-space geometry, density, compactness, neighborhood structure, and representation stability. Using CIFAR-10 as the proposed in-distribution dataset and CIFAR-100, SVHN, and MNIST as candidate OOD datasets, the study evaluates distance-based, density-based, reconstruction-based, and stability-based scores. Detection performance is assessed using the Area Under the Receiver Operating Characteristic Curve (AUROC), Area Under the Precision–Recall Curve (AUPR), and False Positive Rate at 95% True Positive Rate (FPR@95TPR). Statistical analysis is proposed to investigate the relationship between latent-space characteristics and detection effectiveness. Illustrative results are provided to demonstrate the intended reporting format; they are not empirical findings. The framework aims to establish whether measurable representation properties can explain differences in OOD detection across generative model families.**

***Index Terms*—Out-of-distribution detection, generative models, latent-space geometry, variational autoencoders, generative adversarial networks, diffusion models, density estimation, representation learning.**

I. Introduction

Deep learning systems are commonly developed under the assumption that deployment data will resemble their training data. In real-world applications, this assumption may not hold. An industrial inspection system may encounter an unfamiliar defect, a medical imaging model may receive images from a different scanner, or an autonomous system may encounter an object absent from its training dataset. Such inputs are commonly referred to as out-of-distribution (OOD) samples.

OOD detection attempts to distinguish in-distribution (ID) inputs, which are consistent with the reference training distribution, from OOD inputs that exhibit a distributional shift. Early work demonstrated that neural networks can assign high confidence to unfamiliar samples, motivating methods that examine signals beyond ordinary classification confidence [4]. Subsequent approaches have investigated input preprocessing, feature-space distance, and energy-based scores [5], [6], [11].

Generative models provide another direction for this problem. Variational Autoencoders learn probabilistic representations through an encoder–decoder architecture [1]. Generative Adversarial Networks learn a generator through an adversarial objective [2], while Diffusion Models learn to reverse a progressive noise process [3]. Their representations differ in construction and statistical structure, potentially affecting how familiar and unfamiliar samples are separated.

However, a generative model does not automatically provide a reliable OOD detector. Nalisnick *et al.* demonstrated that deep generative models may assign higher likelihoods to certain OOD samples than to ID samples [7]. These observations suggest that OOD detection should not depend on a single signal, such as likelihood or reconstruction error.

Existing research has explored metric-based latent spaces [8] and the behavior of pretrained GAN and diffusion spaces on OOD data [9]. This work proposes a common evaluation framework for studying latent-space geometry and density across these model families. The main objective is to understand which representation characteristics are associated with effective OOD detection, rather than merely identifying the model with the highest detection score.

The intended contributions are:

1) A unified framework for comparing VAE, GAN, and diffusion-based representations under a common ID/OOD evaluation protocol.
2) Quantitative analysis of latent distance, compactness, density, neighborhood structure, and representation stability.
3) Comparison of multiple OOD scoring methods using established detection metrics.
4) Statistical investigation of relationships between representation properties and detection performance.
5) An exploratory composite measure of latent-space OOD awareness, subject to validation on independent datasets.

These contributions are research objectives, not established findings; final claims depend on completed experiments.

II. Literature Survey

A. Generative Models and Latent Representations

Kingma and Welling introduced Auto-Encoding Variational Bayes, establishing a probabilistic framework for learning latent representations [1]. A VAE encodes an input into a latent distribution and reconstructs it through a decoder. Its latent representation and reconstruction behavior provide possible signals for OOD analysis, although regularization does not guarantee ID/OOD separation.

Goodfellow *et al.* introduced Generative Adversarial Networks, which learn a generator and discriminator through adversarial training [2]. A conventional GAN maps latent noise to generated data but does not necessarily encode observed inputs into latent vectors. OOD analysis therefore requires an encoder-equipped GAN or an explicit inversion procedure.

Ho *et al.* introduced Denoising Diffusion Probabilistic Models, which generate samples by progressively reversing a noise process [3]. For OOD analysis, diffusion representations may be extracted from intermediate features or from a latent-diffusion encoder. The selected representation must be defined consistently.

B. Established OOD Detection Methods

Hendrycks and Gimpel introduced a baseline using maximum softmax probability to identify potentially misclassified or OOD samples [4]. Liang *et al.* proposed ODIN, which combines temperature scaling with input preprocessing [5]. These methods provide useful references but depend on classifier outputs rather than directly analyzing generative latent spaces.

Lee *et al.* proposed a Mahalanobis-distance-based framework that models feature distributions and measures deviations from them [6]. Liu *et al.* introduced energy-based OOD detection, using an energy score to distinguish familiar and unfamiliar inputs [11]. Sun *et al.* investigated nearest-neighbor distances in deep feature spaces without requiring a global Gaussian assumption [10]. Cui and Wang reviewed deep-learning-based OOD detection methods and their evaluation practices [12].

C. OOD Detection Using Generative Models

Nalisnick *et al.* showed that deep generative models may assign higher likelihoods to certain OOD datasets than to their training data [7]. Dinari and Freifeld proposed a variational and metric-based latent space designed to improve OOD detection [8]. Ramachandran *et al.* examined pretrained GAN and diffusion models under OOD inversion and reported differences in how their latent spaces represent unfamiliar images [9]. Ran *et al.* investigated VAE-based OOD detection using uncertainty estimation [13].

D. Research Gap

The literature establishes that latent-space structure, feature distances, uncertainty, and generative-model behavior can influence OOD detection. However, the results of an individual detection method do not necessarily explain why a representation works well or whether its advantages generalize across architectures. This study addresses three questions: how geometric and density characteristics differ across VAE, GAN, and diffusion representations; which measurable characteristics are associated with improved OOD detection; and whether these relationships persist across different types of distribution shift.

III. Proposed Work

A. Problem Formulation

Let P_{ID} denote the in-distribution probability distribution and P_{OOD} denote a distribution representing unfamiliar inputs. For model m, define a representation function

$$f_m : \mathcal{X} \to \mathbb{R}^{d_m}, \tag{1}$$

where $\mathcal{X}$ is the input space and d_m is the representation dimension. The representation of an input x is

$$z_m = f_m(x). \tag{2}$$

An OOD score $s_m(x) \in \mathbb{R}$ is defined such that larger values indicate greater evidence of OOD status. A threshold τ produces

$$\hat{y}(x) = \begin{cases} 1, & s_m(x) > \tau, \\ 0, & s_m(x) \le \tau, \end{cases} \tag{3}$$

where 1 denotes OOD and 0 denotes ID. The central objective is to investigate whether measurable characteristics of z_m explain differences in detection performance.

B. Analysis Dimensions

The framework considers five dimensions: (1) latent geometry, including distance and ID–OOD separation; (2) density, estimating how familiar a representation is under the ID reference distribution; (3) compactness, measuring ID spread; (4) neighborhood structure, measuring local distances to ID samples; and (5) representation stability under small, controlled input perturbations. Reconstruction error is included where a meaningful reconstruction procedure is available.

C. Exploratory OOD-Awareness Measure

A composite measure may be constructed from standardized latent-space properties:

$$A_m = \sum_{j=1}^{p} w_j \tilde{u}_{m,j}, \tag{4}$$

where $\tilde{u}_{m,j}$ is the standardized value of property j for model m, and w_j is its weight. Candidate properties include ID–OOD separation, density contrast, neighborhood structure, compactness, and stability. Feature directions must be defined consistently. The measure will be retained only if independent evaluation demonstrates a reproducible relationship with OOD detection performance.

TABLE I
PROPOSED DATASET CONFIGURATION

Dataset	Role	Purpose
CIFAR-10	ID	Model training and reference distribution
CIFAR-100	Related OOD	Evaluate a more challenging shift
SVHN	OOD	Evaluate a different image domain
MNIST	Distinct OOD	Evaluate a visually different distribution

IV. METHODOLOGY

A. Dataset Configuration

CIFAR-10 is selected as the proposed ID dataset because it contains diverse natural-image categories and is widely used in image-learning experiments [15]. CIFAR-100 is considered a related OOD dataset, while SVHN provides a different image domain [14]. MNIST is included as an additional visually distinct evaluation dataset.

The exact splits and sample counts will be fixed before evaluation. Images will be resized and normalized using a common preprocessing pipeline. The ID test set and all OOD test sets will remain separate from model training.

B. Model Configuration

Variational Autoencoder: The VAE uses an encoder $q_\phi(z \mid x)$ and decoder $p_\theta(x \mid z)$. Its objective is

$$\mathcal{L}_{\mathrm{VAE}} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] + \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \parallel p(z)), \tag{5}$$

where the first term measures reconstruction quality and the second regularizes the latent distribution. The encoder posterior mean is used as the deterministic representation:

$$z_{\mathrm{VAE}}(x) = \mu_\phi(x). \tag{6}$$

Generative Adversarial Network: The GAN uses generator G and discriminator D:

$$\min_G \max_D \big[\mathbb{E}_{x\sim P_{\mathrm{ID}}} \log D(x) + \mathbb{E}_{z\sim p(z)} \log(1 - D(G(z)))\big]. \tag{7}$$

Because a standard GAN does not necessarily encode observed inputs, an encoder-equipped GAN or fixed inversion procedure will be used. If an encoder E is selected, $z_{\mathrm{GAN}}(x) = E(x)$. The encoder or inversion method will be documented.

Diffusion Model: A diffusion model gradually adds noise to an input. Its forward process is

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I\right), \tag{8}$$

where t denotes the diffusion timestep and $\bar{\alpha}_t$ is determined by the noise schedule. A standard denoising objective is

$$\mathcal{L}_{\mathrm{diff}} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right]. \tag{9}$$

A fixed intermediate feature representation will be extracted using a documented layer and timestep, since a diffusion model does not inherently provide the same deterministic encoding as a VAE.

C. Latent Geometry and Density

For ID representations $Z_{\mathrm{ID}} = \{z_1, \ldots, z_N\}$, the empirical mean and covariance are

$$\mu_{\mathrm{ID}} = \frac{1}{N}\sum_{i=1}^{N} z_i, \tag{10}$$

$$\Sigma_{\mathrm{ID}} = \frac{1}{N-1}\sum_{i=1}^{N}(z_i - \mu_{\mathrm{ID}})(z_i - \mu_{\mathrm{ID}})^T. \tag{11}$$

The Euclidean score is

$$s_{\mathrm{EUC}}(z) = \|z - \mu_{\mathrm{ID}}\|_2, \tag{12}$$

while the regularized Mahalanobis score is

$$s_{\mathrm{MAH}}(z) = \sqrt{(z - \mu_{\mathrm{ID}})^T(\Sigma_{\mathrm{ID}} + \lambda I)^{-1}(z - \mu_{\mathrm{ID}})}, \tag{13}$$

where λ is a positive regularization parameter.

A Gaussian density estimate provides the negative log-density score

$$s_{\mathrm{DEN}}(z) = -\log p_{\mathrm{ID}}(z). \tag{14}$$

To reduce reliance on a Gaussian assumption, a k-nearest-neighbor score is also considered:

$$s_{\mathrm{kNN}}(z) = r_k(z), \tag{15}$$

where $r_k(z)$ is the distance to the k-th nearest ID reference representation. ID compactness is measured using

$$C_{\mathrm{ID}} = \frac{1}{N}\sum_{i=1}^{N}\|z_i - \mu_{\mathrm{ID}}\|_2^2. \tag{16}$$

The difference between ID and OOD mean representations is

$$S_{\mathrm{mean}} = \|\mu_{\mathrm{ID}} - \mu_{\mathrm{OOD}}\|_2. \tag{17}$$

Raw distances depend on representation scale; therefore, these measurements will be interpreted alongside normalized spread and detection performance.

D. Stability and Reconstruction

For a controlled perturbation $T_\epsilon(x)$, representation stability is measured by

$$s_{\mathrm{STAB}}(x) = \frac{\|f_m(T_\epsilon(x)) - f_m(x)\|_2}{\|f_m(x)\|_2 + \varepsilon}, \tag{18}$$

where ε prevents division by zero. Perturbation magnitude and type will remain fixed across evaluations.

Where reconstruction is available, the reconstruction score is

$$s_{\mathrm{REC}}(x) = \frac{1}{D}\|x - \hat{x}\|_2^2, \tag{19}$$

where D is the number of input elements and $\hat{x}$ is the reconstruction. For GANs and diffusion models, the reconstruction procedure must be specified before comparing this score with the VAE score.

TABLE II
HYPOTHETICAL OOD DETECTION RESULTS (NOT EXPERIMENTAL DATA)

Model	AUROC (%)	AUPR (%)	FPR (%)
VAE	84.2	83.1	38.4
GAN	80.6	79.4	45.2
Diffusion	87.1	86.3	31.8

E. Evaluation Metrics

The framework uses AUROC, AUPR, and FPR@95TPR. With OOD defined as the positive class, AUROC can be interpreted as

$$\mathrm{AUROC} = P(s(x_{\mathrm{OOD}}) > s(x_{\mathrm{ID}})), \tag{20}$$

with standard handling of ties. The true positive rate and false positive rate are

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad \mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}. \tag{21}$$

FPR@95TPR is the false positive rate at the threshold corresponding to 95% true positive rate. Lower values are preferable. AUPR will be reported with OOD as the positive class and with the sample prevalence specified.

F. Statistical Analysis

The association between latent properties and detection performance will be measured using Pearson correlation:

$$r_{uv} = \frac{\sum_{i=1}^{M}(u_i-\bar{u})(v_i-\bar{v})}{\sqrt{\sum_{i=1}^{M}(u_i-\bar{u})^2}\sqrt{\sum_{i=1}^{M}(v_i-\bar{v})^2}}, \tag{22}$$

where u_i denotes a latent property and v_i denotes a detection metric. Spearman correlation will be considered for monotonic nonlinear relationships. Repeated seeds and bootstrap confidence intervals will be used where feasible. Correlation will be interpreted as association, not proof of causation.

V. RESULTS AND ANALYSIS

Important: The numerical values in Tables II and III are hypothetical examples created only to demonstrate presentation and analysis. They are not measured results from trained models and must be replaced with actual experimental outputs before submission.

A. Illustrative Overall Performance

Assume, for demonstration only, that each model is evaluated on the same ID test set and three OOD datasets using a predefined latent-distance score.

Under this hypothetical example, the diffusion representation has the highest mean AUROC and AUPR and the lowest FPR@95TPR. These values demonstrate metric interpretation, not a finding about actual model performance. Real experiments may produce a different ranking across OOD datasets.

TABLE III
HYPOTHETICAL NORMALIZED LATENT-SPACE INDICATORS (NOT EXPERIMENTAL DATA)

Model	Compactness	Separation	Density Contrast
VAE	0.72	0.68	0.65
GAN	0.61	0.57	0.59
Diffusion	0.75	0.79	0.76

B. Illustrative Latent-Space Measurements

These synthetic values assume that larger compactness values indicate greater concentration, while larger separation and density-contrast values indicate stronger distinction between ID and OOD representations. In this hypothetical scenario, diffusion representations exhibit stronger separation and density contrast. If observed experimentally, such a pattern would warrant further investigation; it cannot be inferred from these illustrative numbers.

C. Required Experimental Analysis

The completed study will report results separately for CIFAR-100, SVHN, and MNIST; compare distance, density, reconstruction, and stability scores where applicable; report variation across random seeds and confidence intervals; evaluate correlations between latent-space properties and detection metrics; and assess whether the findings persist on an additional OOD dataset. The composite measure will be evaluated separately from the data used to select its weights. If it fails to show a stable relationship with detection performance, it will not be presented as a validated contribution.

VI. CONCLUSION

This paper proposes a unified framework for examining how latent-space geometry and density relate to OOD detection across VAEs, GANs, and Diffusion Models. The framework combines distance-based and density-based scores with measurements of compactness, neighborhood structure, representation stability, and reconstruction behavior.

By evaluating the models under a common dataset protocol and using AUROC, AUPR, and FPR@95TPR, the study aims to identify which latent-space properties are associated with effective detection. The analysis also considers whether a composite measure can summarize these properties without losing important architectural or dataset-specific information.

The framework does not assume that one generative model is inherently more OOD-aware than another. Its purpose is to investigate the conditions under which representation characteristics are useful for distinguishing familiar from unfamiliar data. The illustrative results in this draft are placeholders and cannot support scientific conclusions. Final claims must be based on trained models, recorded measurements, and statistical validation.

Future work may extend the evaluation to larger datasets and additional generative architectures. Another direction is to use validated latent-space findings to design a training objective that directly improves OOD separation, moving the research

from comparative analysis toward a method for constructing more OOD-aware representations.

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